Cadra Mobile — AI Item Identification & Referencing

Requirements addendum to the Cadra Barcode Station mobile app BRD · Oct 13, 2026 · @soji

1. Purpose and scope

The Barcode Station mobile app gains an AI capture mode that photographs an item with no manufacturer ID, proposes its asset class and attributes, and registers it under a Cadra-issued unique reference in one pass. It targets furniture and fittings (chairs, tables, desks, cabinets, shelving, sofas, partitions) and any other item with no serial, model plate or chassis number.

This addendum extends the Barcode Station mobile app BRD and serves three award lots:

- **Lot 1 (asset management solution):** AI-assisted registration in the asset register.
- **Lot 4 (audit and tagging of existing assets):** the main use case. Field teams walk rooms, capture untagged items, and tag them on the spot.
- **Lot 6 (onsite Barcode Stations):** the same capture flow at regional and state office stations.

Out of scope: items that already carry a reliable serial or chassis number (laptops, vehicles, servers). These follow the existing capture flow, where the asset class sets the required fields.

2. Design principles

The AI proposes and a person confirms; Cadra alone issues the reference. These rules keep the feature consistent with decisions already made for the tag and register design.

1. **Cadra issues the reference, not the model.** The unique reference is the standard Cadra asset code from the register sequence. The model never invents or derives an identifier.
2. **The code stays opaque and immutable.** As already decided, the barcode and QR carry the asset code verbatim. No AI-predicted value (class, colour, location) is encoded in it, because predictions can be corrected later.

3. **A person confirms every proposal.** The model pre-fills class and attributes. The operator accepts or corrects each one before the record is saved.
4. **The photo record substitutes for a serial number.** For items with no manufacturer ID, the reference photos, visual signature and capture context together become the evidence of identity.
5. **Classification feeds the existing cascade.** AI output maps to the Cadra asset class catalog. It acts as a third source after the PO line and receipt confirmation, and never bypasses them.
6. **It works offline.** Capture, classification and code reservation must work without network access at regional offices, then sync.

3. Functional requirements

Fourteen requirements cover capture, classification, referencing, duplicate control and re-identification; ten are Must for the Lot 4 audit.

ID	Requirement	Priority
AI-FR-01	**Guided capture.** The app prompts for a minimum photo set per item: front, side, and a close-up of any distinguishing mark (scratch, repair, sticker, maker's stamp). It rejects blurred or dark shots before saving.	Must
AI-FR-02	**Class suggestion.** The model proposes the asset class and sub-type from the catalog (e.g. Furniture > Seating > Executive chair) with a confidence score. It shows the top 3 when confidence is below the threshold.	Must
AI-FR-03	**Attribute extraction.** The model pre-fills class-specific attributes: material, colour, approximate dimensions band, upholstery, number of drawers or seats, brand where a logo or stamp is visible.	Must
AI-FR-04	**Condition grade.** The model proposes a condition grade (New, Good, Fair, Poor, Unserviceable) and flags visible damage. The operator confirms it.	Should

ID	Requirement	Priority
AI-FR-05	**Operator confirmation.** Every AI value is shown as a suggestion. The record saves only after the operator accepts or edits each mandatory field.	Must
AI-FR-06	**Reference issue.** On confirmation, the app takes the next asset code from a block of codes reserved to that device, so it works offline. It then prints the tag at the station or queues it.	Must
AI-FR-07	**Visual signature.** The app computes an image embedding from the photo set and stores it against the asset as its visual signature.	Must
AI-FR-08	**Duplicate check.** Before issuing a code, the app compares the new signature with registered assets at the same location and nearby. A close match triggers a "possible duplicate" prompt showing the existing record.	Must
AI-FR-09	**Bulk room mode.** The operator photographs a room. The model detects and counts items by class (e.g. 12 chairs, 3 tables) and creates draft records for confirmation and batch tag printing.	Must
AI-FR-10	**Room context.** Each capture records the scanned room or zone barcode, GPS where available, operator, device and time. Location stays a register attribute, never part of the code.	Must
AI-FR-11	**Lost-tag recovery.** If a tag is missing, the operator photographs the item. The app ranks likely matches from that location by visual signature. Confirming a match prints a replacement tag with the same asset code.	Must
AI-FR-12	**Audit verification.** During audits, scanning a tag and taking one photo checks the item against its stored photos. A mismatch flags a possible tag swap for review.	Should

ID	Requirement	Priority
AI-FR-13	**Low-value pooling.** For items below the NRS capitalisation threshold, the operator can register a pooled record (e.g. "Conference room B chairs, qty 12") under one code instead of 12.	Should
AI-FR-14	**Feedback loop.** Every operator correction is stored as labelled training data for the next model version.	Should

4. Reference generation and identical items

The tag is the identity; the AI record is the evidence. Mass-produced furniture is often visually identical, so no model can tell twelve matching chairs apart once their tags are gone. The design must not rely on that.

Reference rules

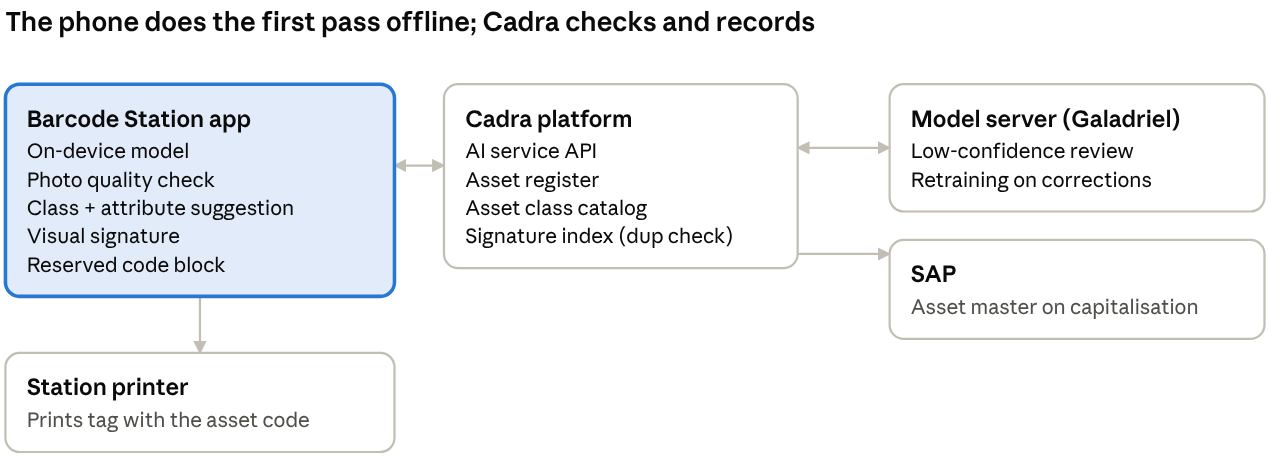
- The reference is the standard Cadra asset code from the register sequence, with the same format as all other classes. AI-registered items get no separate scheme.
- Each device holds a reserved block of codes for offline use. Unused codes return to the pool on sync. A code is never reused, even after the asset is disposed of.
- The record stores how the item was identified: `AI-assisted`, plus the model version and the operator who confirmed it.

Handling identical items

- **Distinguishing marks first.** Capture prompts for any wear, repair, sticker or stamp. These make the visual signature useful for re-identification.
- **Location narrows the search.** Lost-tag matching and duplicate checks are scoped to the item's room or zone, where only a few candidates exist.
- **Tamper-evident tags carry the load.** For identical items, re-identification is a ranked suggestion, not proof. A missing tag on an identical item is resolved by the custodian and supervisor, not by the model alone.
- **Pooling where policy allows.** Low-value identical sets can be one pooled record (AI-FR-13), which removes the problem for those items.

5. Model and architecture

A small model runs on the phone for instant, offline suggestions; a larger server model handles low-confidence cases and retraining.



AI capture architecture · phone, Cadra, model server, SAP

The phone never needs the server to register an item. It syncs records, photos and signatures when a connection returns.

- **On-device model:** a compact object detector (for bulk room counts) plus a classifier and embedding model. It must run on mid-range Android devices issued to field teams.
- **Server model:** hosted on WML's own ML platform (Galadriel). It re-scores low-confidence captures after sync and is retrained on operator corrections.
- **Starting point:** pretrained vision models fine-tuned on NRS's own furniture catalog. The first Lot 4 pilot sites produce the initial labelled dataset.
- **Model versioning:** each record stores the model version that made the suggestion. A new version ships only after it beats the current one on a held-out NRS test set.
- **Catalog mapping:** model labels map one-to-one to Cadra asset class codes. A class added to the catalog is unavailable to the model until it is retrained.
- **SAP:** AI-registered items follow the normal capitalisation rule. Classification gaps block capitalisation, never registration.

6. Non-functional, privacy and audit requirements

Every AI suggestion must be traceable, and photos must not become a record of people.

ID	Requirement	Priority
AI-NFR-01	**Speed.** On-device suggestion within 2 seconds of the last photo; bulk room count within 5 seconds.	Must
AI-NFR-02	**Offline.** Full capture, suggestion and code issue with no network for at least 5 working days of fieldwork, then automatic sync.	Must
AI-NFR-03	**Privacy (NDPA 2023).** Faces and screens in the background are detected and blurred on the device before photos are stored or synced.	Must
AI-NFR-04	**Data residency.** Photos, signatures and training data are stored in NRS-approved hosting in Nigeria. They are not sent to third-party AI APIs.	Must
AI-NFR-05	**Audit trail.** Each record keeps the AI suggestion, the final confirmed values, which fields were changed, the operator, the model version and timestamps.	Must
AI-NFR-06	**Photo retention.** Reference photos are kept for the life of the asset plus the NRS retention period after disposal.	Must
AI-NFR-07	**Storage.** Photos are compressed to a maximum of 300 KB each on the device before sync.	Should
AI-NFR-08	**Model monitoring.** A dashboard shows suggestion acceptance rate by class, site and model version, so weak classes are retrained first.	Should

7. Acceptance criteria and open questions

The feature is accepted when it meets these targets on an NRS pilot site, measured against a manually verified count.

Measure	Target
Top-1 class accuracy, furniture sub-types	at least 90%

Measure	Target
Correct class in top 3 suggestions	at least 98%
Bulk room count error per class	within ±1 item per room
Duplicate check catches re-captured items	at least 95%
Lost-tag recovery: true match in top 3, non-identical items	at least 90%
Average capture time per item, AI mode vs manual	at least 40% faster

These targets are proposed starting points and should be confirmed after the pilot dataset is labelled.

Open questions for NRS

- [ ] What is the capitalisation threshold for furniture and fittings, and may items below it be pooled (AI-FR-13)?
- [ ] Which pilot sites should Lot 4 start at to build the training dataset?
- [ ] Which device models will field teams use? This sets the on-device model size.
- [ ] Does NRS want condition grades (AI-FR-04) to drive maintenance or disposal workflows?
- [ ] Who approves a lost-tag replacement for identical items: custodian, supervisor, or both?
- [ ] Is the photo retention period after disposal set by NRS records policy?